ITDS 124 Mathematics of Computing

Chapter 1. Fundamentals

1.4 (Part 1) Properties of Integers

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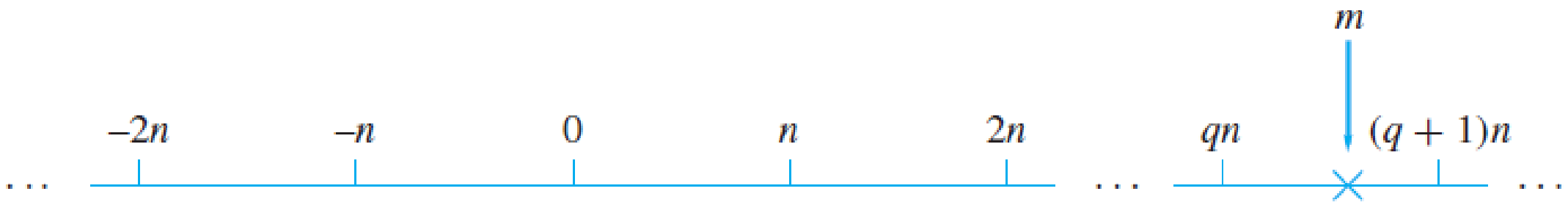
Theorem 1

- If n and m are integers and $n > 0$, we can write $m = qn + r$ for integers q and r with $0 \leq r < n$. Moreover, there is just one way to do this.



Examples

- If n is 3 and m is 16, then $16 = 5(3) + 1$ so q is 5 and r is 1.
- If n is 10 and m is 3, then $3 = 0(10) + 3$ so q is 0 and r is 3.
- If n is 5 and m is -11, then $-11 = -3(5) + 4$ so q is -3 and r is 4.

Divisibility

- If the r in Theorem 1 is zero, so that m is a multiple of n, we write $n \mid m$, which is read "n divides m."
- If $n \mid m$, then $m = qn$ and $n \leq |m|$.
- If m is not a multiple of n, we write $n \nmid m$, which is read "n does not divide m."

- Examples : $4 \mid 8$, $3 \mid 27$, $11 \mid 132$

  $4 \nmid 9$, $3 \nmid 25$, $11 \nmid 25$

Theorem 2

- Let a, b, and c be integers.
  a) If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$.
  b) If $a \mid b$ and $a \mid c$, where $b > c$, then $a \mid (b - c)$.
  c) If $a \mid b$ or $a \mid c$, then $a \mid bc$.
  d) If $a \mid b$ and $b \mid c$, then $a \mid c$.

Prime numbers

- A number p is called **prime** if the only positive integers that divide p are p and 1.
- Examples of prime numbers: 2, 3, 5, 7, 11, and 13.
- There is a science to determining prime numbers.
- The first 30 prime numbers are 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, 101, 103, 107, 109, 113.
- The following slides present some computer algorithms that can be used to determine if a number $n > 1$ is prime.

(1) Basic prime number algorithm

To check whether n is prime number

1. First, check if $n = 2$. If it is, n is prime. Otherwise, proceed to step 2.

2. Check to see if each integer k is a divisor of n where $1 < k < (n - 1)$. If none of the values of k are divisors of n, then n is prime.

(2) Better prime number algorithm

Note that if $n = mk$, then either m or k is less than $\sqrt{n}$. Therefore, we don't need to check for values of $k > \sqrt{n}$.

1. First check if $n = 2$. If it is, n is prime. Otherwise, proceed to step 2.

2. Check to see if each integer k is a divisor of n where $1 < k < \sqrt{n}$. If none of the values of k are divisors of n, then n is prime.

(3) Even better prime number algorithm

Note that if $k \mid n$, and k is even, then $2 \mid n$. Therefore, if 2 does not divide n, then no even number can be a divisor of n. (If $a \mid b$ and $b \mid c$, then $a \mid c$)

1. First check if $n = 2$. If it is, n is prime. Otherwise, proceed to step 2.

2. Check if $2 \mid n$. If so, n is not prime. Otherwise, proceed to step 3.

3. Check to see if each odd integer k is a divisor of n where $1 < k < \sqrt{n}$. If none of the values of k are divisors of n, then n is prime.

(4) Even^2 better prime number algorithm

Note that if $d \mid k$ and $k \mid n$, then $d \mid n$. Therefore, if d does not divide n, then no multiple of d can be a divisor of n.

1. First check if $n = 2$. If it is, n is prime. Otherwise, proceed to step 2.

2. Use a sequence $k = 2,3,5,7,11,13,17, \ldots$ up to $\sqrt{n}$ to check if $k \mid n$. If no values of k are divisors of n, then n is prime. (Note that list is a list of prime numbers !)

Theorem 3

- Every positive integer $n > 1$ can be written uniquely as $p_1^{k_1} p_2^{k_2} \ldots p_s^{k_s}$, where $p_1 < p_2 < \cdots < p_s$ are distinct primes that divide n and the k's are positive integers giving the number of times each prime occurs as a factor of n.
- Examples
  - $9 = 3 * 3 = 3^2$
  - $24 = 8 * 3 = 2 * 2 * 2 * 3 = 2^3 * 3$
  - $30 = 2 * 3 * 5$
- Basically, this means that any number can be broken into multiples of prime numbers.

Greatest Common Divisor (GCD) (ตัวหารร่วมมากสุด)

- If a, b, and k are in Z^+, and $k \mid a$ and $k \mid b$, then k is a common divisor of a and b.
- If d is the largest such k, then d is called the greatest common divisor (GCD) of a and b, and that means $d = GCD(a, b)$.
- Example : If $a = 36, b = 24$
  - Common divisors of a and b are 1,2,4,6,12
  - The greatest common divisor of a and b is 12.
  - $GCD(36,24) = 12$

Theorem 4

If d is $GCD(a, b)$, then

a) $d = sa + tb$ for some integers s and t.
(s and t can be either positive or negative)

b) If c is any other common divisor of a and b, then $c \mid d$.

Examples

- $GCD(30,12) = 6$ ➔ $6 = s * 30 + t * 12 = 1 * 30 + (-2) * 12$
- $GCD(95,17) = 1$ ➔ $1 = s * 95 + t * 17 = (-5) * 95 + 28 * 15$

(Note that 17 is prime and $17 \nmid 95$)

Euclidean algorithm for finding $GCD(a, b)$ (1)

Suppose that $a > b > 0$. Then by Theorem 1, we can write

$$a = k_1 b + r_1, \quad \text{where } k_1 \text{ is in } \mathbb{Z}^+ \text{ and } 0 \leq r_1 < b.$$

From Theorem 2, it tells us that if n divides a and b, then it must divide r_1, since $r_1 = a - k_1 b$.

Similarly, if n divides b and r_1, then it must divide a.

Therefore, we can see that the common divisors of a and b are the same as the common divisors of b and r_1, so $GCD(a, b) = GCD(b, r_1)$.

Euclidean algorithm for finding $GCD(a, b)$ (2)

Now continue using Theorem 1 as follows:

	$a = k_1 b + r_1$	$0 \le r_1 < b$
Divide b by r_1 :	$b = k_2 r_1 + r_2$	$0 \le r_2 < r_1$
Divide r_1 by r_2 :	$r_1 = k_3 r_2 + r_3$	$0 \le r_3 < r_2$
Divide r_2 by r_3 :	$r_2 = k_4 r_3 + r_4$	$0 \le r_4 < r_3$
$\vdots$	$\vdots$	$\vdots$
Divide r_{n-2} by r_{n-1} :	$r_{n-2} = k_n r_{n-1} + r_n$	$0 \le r_n < r_{n-1}$
Divide r_{n-1} by r_n :	$r_{n-1} = k_{n+1} r_n + r_{n+1}$	$0 \le r_{n+1} < r_n$

Since $a > b > r_1 > r_2 > r_3 > r_4 > \cdots$, the remainder will eventually become zero, so at some point we have $r_{n+1} = 0$.

Euclidean algorithm for finding $GCD(a, b)$ (3)

Previously, we know that $GCD(a, b) = GCD(b, r_1)$.

Repeating this argument with b and r_1 ➔ $GCD(b, r_1) = GCD(r_1, r_2)$.

Similarly, $GCD(a, b) = GCD(b, r_1) = GCD(r_1, r_2) = \cdots = GCD(r_{n-1}, r_n)$.

Since $r_{n-1} = k_{n+1} r_n$ ➔ $GCD(r_{n-1}, r_n) = r_n$.

Hence $r_n = GCD(a, b)$

Example :

Compute $\boldsymbol{GCD(273, 98)}$

Let $a = 273, \quad b = 98.$ Using the Euclidean algorithm,

Divide 273 by 98 :	$273 = 2 * 98 + 77$
Divide 98 by 77 :	$98 = 1 * 77 + 21$
Divide 77 by 21 :	$77 = 3 * 21 + 14$
Divide 21 by 14 :	$21 = 1 * 14 + 7$
Divide 14 by 7 :	$14 = 2 * 7 + 0$

So $GCD(273,98) = 7,$ the last of the nonzero divisors.

Finding s, t in $d = GCD(a, b) = sa + tb$ (1)

- In theorem 4(a), we know that $d = GCD(a, b)$, we can find integers s and t such that $d = sa + tb$. Finding s, t can be found as follows.
- From the next-to-last equation in the Euclidean algorithm,

  $r_{n-2} = k_n r_{n-1} + r_n$, we can rewrite as $r_n = r_{n-2} - k_n r_{n-1}$.
- Next solve the second-to-last equation in the Euclidean algorithm,

  $r_{n-3} = k_{n-1} r_{n-2} + r_{n-1}$, we can rewrite as $r_{n-1} = r_{n-3} - k_{n-1} r_{n-2}$.
- And we can substitute r_{n-1} in the r_n equation, we then have

  $r_n = r_{n-2} - k_n[r_{n-3} - k_{n-1} r_{n-2}]$.
- Continue to work up through the equations in the Euclidean algorithm, replacing r_i by an expression involving r_{i-1} and r_{i-2}, and finally arriving at an expression involving only a and b.

Finding s, t in $d = GCD(a, b) = sa + tb$ (1)

	$a = k_1 b + r_1$	$r_1 = a - k_1 b$
Divide b by r_1 :	$b = k_2 r_1 + r_2$	$r_2 = b - k_2 r_1$
Divide r_1 by r_2 :	$r_1 = k_3 r_2 + r_3$	$r_3 = r_1 - k_3 r_2$
Divide r_2 by r_3 :	$r_2 = k_4 r_3 + r_4$	$r_4 = r_2 - k_4 r_3$
$\vdots$	$\vdots$	$\vdots$
Divide r_{n-4} by r_{n-3} :	$r_{n-4} = k_{n-2} r_{n-3} + r_{n-2}$	$r_{n-2} = r_{n-4} - k_{n-2} r_{n-3}$
Divide r_{n-3} by r_{n-2} :	$r_{n-3} = k_{n-1} r_{n-2} + r_{n-1}$	$r_{n-1} = r_{n-3} - k_{n-1} r_{n-2}$
Divide r_{n-2} by r_{n-1} :	$r_{n-2} = k_n r_{n-1} + r_n$	$r_n = r_{n-2} - k_n r_{n-1}$
Divide r_{n-1} by r_n :	$r_{n-1} = k_{n+1} r_n + r_{n+1}$	

Example: Finding s, t in $GCD(273, 98) = s(273) + t(98)$ (1)

First, compute $\boldsymbol{GCD(273, 98)}$

Let $a = 273, \quad b = 98.$ Using the Euclidean algorithm,

Divide 273 by 98 :	$273 = 2 * 98 + 77$
Divide 98 by 77 :	$98 = 1 * 77 + 21$
Divide 77 by 21 :	$77 = 3 * 21 + 14$
Divide 21 by 14 :	$21 = 1 * 14 + 7$
Divide 14 by 7 :	$14 = 2 * 7 + 0$

Example: Finding s, t in $GCD(273, 98) = s(273) + t(98)$ (2)

- Next, from rewrite the next-to-last equation in the Euclidean algorithm, and rewrite the second-to-last equation in the Euclidean algorithm and substitute with the previous equation

Divide 273 by 98 :	$273 = 2 * 98 + 77$	
Divide 98 by 77 :	$98 = 1 * 77 + 21$	
Divide 77 by 21 :	$77 = 3 * 21 + 14$	$14 = 77 - 3 * 21$
Divide 21 by 14 :	$21 = 1 * 14 + 7$	$7 = 21 - 1 * 14$
Divide 14 by 7 :	$14 = 2 * 7 + 0$	$= 21 - 1 * (77 - 3 * 21)$
		$= 4 * 21 - 1 * 77$

Example: Finding s, t in $GCD(273, 98) = s(273) + t(98)$ (3)

Next, continue to work up through the equations in the Euclidean algorithm

Divide 273 by 98 :	$273 = 2 * 98 + 77$	
Divide 98 by 77 :	$98 = 1 * 77 + 21$	$21 = 98 - 1 * 77$
Divide 77 by 21 :	$77 = 3 * 21 + 14$	
Divide 21 by 14 :	$21 = 1 * 14 + 7$	$7 = 4 * 21 - 1 * 77$
Divide 14 by 7 :	$14 = 2 * 7 + 0$	$= 4 * (98 - 1 * 77) - 1 * 77$
		$= 4 * 98 - 5 * 77$

Example: Finding s, t in $GCD(273, 98) = s(273) + t(98)$ (4)

Next, continue to work up through the equations in the Euclidean algorithm

Divide 273 by 98 :	$273 = 2 * 98 + 77$	$77 = 273 - 2 * 98$
Divide 98 by 77 :	$98 = 1 * 77 + 21$	
Divide 77 by 21 :	$77 = 3 * 21 + 14$	
Divide 21 by 14 :	$21 = 1 * 14 + 7$	$7 = 4 * 98 - 5 * 77$
Divide 14 by 7 :	$14 = 2 * 7 + 0$	$= 4 * 98 - 5 * (273 - 2 * 98)$
		$= 14 * 98 - 5 * 273$
		$= -5 * 273 + 14 * 98$

So, $s = -5$ and $t = 14$

Example: Finding s, t in $GCD(108, 60) = s(108) + t(60)$ (1)

First, compute $\boldsymbol{GCD(108, 60)}$

Let $a = 108, \quad b = 60.$ Using the Euclidean algorithm,

Divide 108 by 60 : $108 = 1 * 60 + 48$

Divide 60 by 48 : $60 = 1 * 48 + 12$

Divide 48 by 12 : $48 = 4 * 12 + 0$

So $\boldsymbol{GCD(108, 60) = 12}$

Next, to find s, t

$12 = 60 - 1 * 48$ and $48 = 108 - 1 * 60$

$12 = 60 - 1 * (108 - 1 * 60)$

$12 = 2 * 60 - 1 * 108$

So, $\boldsymbol{s = -1, \quad t = 2}$

Theorem 5

- If a and b are in Z^+, $b > a$, then $GCD(a, b) = GCD(b, b \pm a)$.

Example : $GCD(60,108) = 12 = GCD(108,48) = GCD(108,156)$

Least Common Multiple (LCM) (ตัวคูณร่วมน้อยสุด)

- If a, b, and k are in Z^+, and $a \mid k$ and $b \mid k$, then k is a common multiple of a and b.
- If c is the smallest such k, then c is called the least common multiple (LCM) of a and b, and that means $c = LCM(a, b)$.
- Example : If $a = 36, b = 24$
  - Common multiples of a and b are $72,144, \ldots$
  - The least common multiple of a and b is 72.
  - $LCM(36,24) = 72$

Theorem 6

- If a and b are two positive integers, then $GCD(a,b) * LCM(a,b) = ab$.

Proof. Let $p_1, p_2, \dots, p_k$ be all the prime factors of either a or b. Then we can write

$$a = p_1^{a_1} p_2^{a_2} \dots p_k^{a_k} \text{ and } b = p_1^{b_1} p_2^{b_2} \dots p_k^{b_k}$$

Where some of the a_i and b_i may be zero. It then follows that

$$GCD(a,b) = p_1^{\min(a_1,b_1)} p_2^{\min(a_2,b_2)} \dots p_k^{\min(a_k,b_k)}$$

And

$$LCM(a,b) = p_1^{max(a_1,b_1)} p_2^{max(a_2,b_2)} \dots p_k^{max(a_k,b_k)}$$

Hence

$$GCD(a,b) * LCM(a,b) = p_1^{a_1+b_1} p_2^{a_2+b_2} \dots p_k^{a_k+b_k}$$

$$= \left(p_1^{a_1} p_2^{a_2} \dots p_k^{a_k}\right) * \left(p_1^{b_1} p_2^{b_2} \dots p_k^{b_k}\right)$$

$$= ab$$

Example

- Let $a = 540$ and $b = 504$. Factoring a and b into primes.

$$a = 540 = 2^2 * 3^3 * 5 \qquad b = 504 = 2^3 * 3^2 * 7$$

- Then we have $p_1 = 2, p_2 = 3, p_3 = 5, p_4 = 7$ as the prime factors of either a or b.

$$a = 540 = 2^2 * 3^3 * 5^1 * 7^0 \qquad b = 504 = 2^3 * 3^2 * 5^0 * 7^1$$

- $\boldsymbol{GCD(540, 504) = 2^{\min(2,3)} * 3^{\min(3,2)} * 5^{\min(1,0)} * 7^{\min(0,1)}}$

$$= 2^2 * 3^2 * 5^0 * 7^0 = 2^2 * 3^2 = 36$$

- $\boldsymbol{LCM(540, 504) = 2^{max(2,3)} * 3^{max(3,2)} * 5^{max(1,0)} * 7^{max(0,1)}}$

$$= 2^3 * 3^3 * 5^1 * 7^1 = 7{,}560$$

- $\boldsymbol{GCD(540, 504) * \mathrm{LCM}(540, 504) = 36 * 7{,}560 = 272{,}160 = 540 * 504}$

Quotient (ผลหาร) and remainder (เศษ, ส่วนที่เหลือ)

- If m and n are integers and $n > 1$, we can write $m = qn + r, \quad 0 \leq r < n.$

q ➔ quotient $\qquad$ r ➔ remainder

- Example 1 : If the time is now 4 o'clock, what time will it be 101 hours from now?
  - Let $n = 12$ and $m = 4 + 101 = 105$. Then $105 = 8 * 12 + 9$. So the time = 9 o'clock.

- Example 2 : The mod functions $\quad f_3(14) = 2 \quad$ ➔ $\quad 14 = 4 * 3 + 2$

  $f_7(153) = 6 \quad$ ➔ $\quad 153 = 21 * 7 + 6$